Day-9

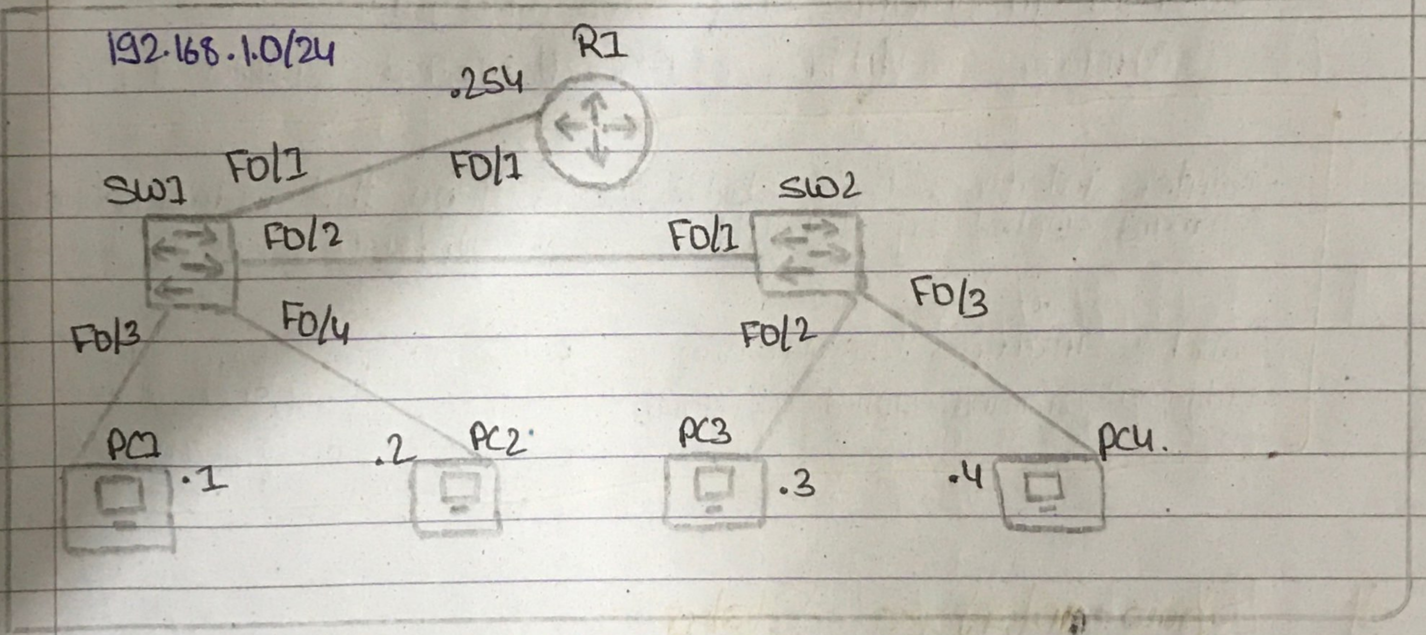
October-10&11

Switch Interfaces.

Specifically, we're going to talk about configuring Layer 1 characteristics of these interfaces, such as speed & duplex.

Speed: It refers to data rate in bits per second, such as 10, 100, or 1000 megabits per second.

Duplex- It refers to whether a device can send & receive data at the same time.



# Let's configure switch 1.

1) Enter priveleged exec mode: Use 'en' or 'enable' command.
2) Then use, 'show ip interface brief'
3) The four interfaces which are connected to devices have up in Status (layer 1) column & protocol (Layer 2) column.

We usually call this up/up. We haven't done any config on SW1 yet. We can see interfaces which aren't connected to devices have down in status column unlike routers which used to have 'administratively down' for all connected & not connected devices until they were configured to up.

3) ~~I~~IP address are unassigned and it will remain this way because these are layer 2 switch ports & they don't need IP address.

(The concept of assigning IP to switches will be done in future)

4) ~~V~~ Interface भी 'VLAN1' दिखते हैं, which too, we would ignore & study in future lesson

5) 'Down/Down' is different than administratively 'down/down'

'Show ip interface brief' -> 6 col -> ~~inter~~

a) interfa ~~e~~ (b) IP c) OK d) Method e) Status f) Protocol

Router interfaces have shutdown command applied by default = will be the administratively down/down state by default

Switches interfaces do NOT have 'shutdown' command applied by default = will be in the up/up state if connected to other devices OR In the down/down state if not connected to other device.

Show interface Status

It has ~~6 to~~ 7-columns.

a) Port: It simply lists interfaces
b) Name: Description of interface
c) Status: Different than 'Status' of 'show ip int bri'. It simply shows whether ~~po~~ int is connected to other device or not! Connected device shows 'connected' & devices which aren't connected shows 'not connect'

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d) VLAN - They are used to divide LAN into smaller LAN, the default ~~a~~ VLAN is 1. Separate vdo will come for this.
(The 'int' connected to SW2 is trunk in our case all other are 1)

~~e)~~

e) duplex - shows whether device can send & receive data at the same time. If it can it's 'full-duplex' if can't 'half-duplex'.
Duplex is auto by default ~~by~~ in CISCO switches, it will negotiate with neighbouring device & use full duplex if possible.
(a-full - automatically negotiated duplex of ~~[illegible]~~ full with ~~nei~~ neighbouring device).

~~f) Speed.~~

f) Speed - They are also ~~default by itself~~ auto by default:
They negotiate with neighbouring device & use fastest speed both devices are capable of.
(a-100 means automatically negotiated for 100 mbps)

g Type - RJ45 होकि SFP होकि त्यो सब देखिन्छ यसमा चाँहि
(10/100 Base Tx - 10/100 Mbps जसमा यो device काम गर्न सक्छ)

Configuring Interface speed & duplex

1) Enter global config mode.
2) Then Same as in router ~~or~~ use '**interface f0/1**' (i.e / or 'int f/g 0/0/1/2')
3) ~~Speed~~ ~~10~~ To configure speed use the command '**speed 10**' ie **speed your-desired&valid-speed**'. You can also choose auto

4) To configure duplex use the command '**duplex your-prefered-mode**'
Choose जुनको mode should be supported by both connected devices.

5. To configure descriptio(i-e name column) do same as in router 'description your-description'

6) If we want common settings for many interfaces, i.e for ~~devices~~ interface which aren't connected

interface range f0/5-12 (interface 5-12).
यसरी केहि apply गर्दा भने sub f0/5- f0/12 सम्म यहि settings use hnxan.

7) interface f0/5-6, f0/9-12.
Consecutively नै select गर्नु पर्छ भन्ने हैन। So, 6, 9, 10, 11, 12 मात्र use गर्नेको cmmnd ley select गर्न मिल्छ।

Full/Half duplex

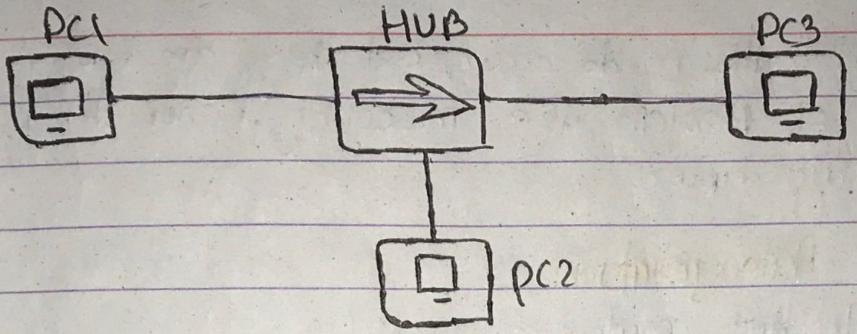
Full duplex - They can send & receive data at the same time. It doesn't have to wait.

Half-duplex - The device cannot send & receive ~~d~~ data at the same time. If it is receiving a frame, it must wait before sending a frame.

{Full duplex shall be preferred). In modern networks that uses switches, all devices can use full duplex on their interfaces}.

In modern-day network, half duplex is used almost nowhere.
To know, where half duplex was used lets see old network:

Switches are more sophisticated than hubs. Hub लाई switch ले Replace गर्यो भने त्यसको network will have 3 collision of domain instead of one.



HUB- Much simpler than switch, it's simply a repeater. Any frame it receives, it floods like a switch does with a broadcast or unknown unicast frame.

If PC1 ~~sends~~ wants to send frame to PC2, HUB receives it floods it. PC3 ignores due to MAC Address incorrect but PC2 accepts it.

What if PC1 & PC3 wants to send frame to PC2 & PC1 respectively at same time. They both send frame out of their interface. Now problem occurs. Hub won't send one first & other later. It simply tries to flood both at same time & this result on collision on interface & PC2 won't receive either frame intact. All device connected to hub are part of collision domain. ~~They~~

To deal with half-duplex situation like this, ethernet devices use mechanism called CSMA/CD.

Carrier-Sense Multiple Access with Collision Detection (CSMA/CD)

It describes how devices avoid collision in a half-duplex situation & how they react if collision do occurs.

a) Before sending frames, devices 'listen' to collision domain until they detect other devices are not sending.

b) If collision occurs, device send a jamming signal to inform other devices that collision ~~occur~~ has happened.

c). Each device will wait a random period of time before sending frames again.

d) The process repeats.

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Switches मा नि collision do occur but they are rare & usually a sign of problem like misconfig, rather than regular occurence in half-duplex.

Speed / Duplex Autonegotiation

- It applies to both routers & switches.

- Interfaces that can run at dffrnt speed (10/100 or 10/100/1000) have default settings of speed auto & duplex auto.

- Interface advertise their capabilities to neighbouring devices & negotiate best speed & duplex setting they are both capable of.

What if autonegotiation is disabled on device cnnctd to switch?

SPEED: The switch will try to sense the speed that the other device is operating at.
If it fails to sense the speed, it will use the slowest supported speed (i.e 10 Mbps on a 10/100/100 int

DUPLEX: If speed is 10/100 Mbps, the switch will use half-duplex.
If speed is 1000 Mbps or greater, it will use full duplex.

Interface errors

Use ~~'intef~~ 'show interfaces f0/2'

Runts: Frames that are smaller than minimum frame size (64 bytes).

CRC - Cyclic Redundancy check. It is used to detect error.

Giants: Frames that are larger than maximum frame size (1518 bytes)

CRC: Frames that failed CRC check (in Ethernet FCS trailer)
Frame: ~~CRC~~ = Frame that have incorrect or illegal format
Input error. Total of various counters like above 4 (Runts, Giants, etc)
Output error: Counts frame the switch tried to send but failed due to an error.

These errors are same on router too.

QUIZ.

1). There is a duplex mis-match between SW1's F0/1 interface and SW2's F0/1 interface, which are connected. Autonegotiation is disabled what will be result.
a) Improved performance [✓ b) Collision will occur.]
c). SW1 will sense SW2 duplex setting & adjust to ~~watch~~ match

2) What is used on ~~a~~ half-duplex interface to detect & avoid collisions?
[✓ a) CSMA/CD.] b) CSMA/CA c) Auto-negotiation d) Duplex-aut

3) Which command shows various counters of errors detected on interface?
[✓ a) show interfaces] b) show ip int bri c) sho ~~w~~ int status d) sh int err

4) Which are eg of errors that might occur on a network interface?
a) Runts, Giants, Brodcasts. b) Shorts, Longs, Oversize, c) Pack, Byts, In, out
[✓ d) Runts, Giants, CRC]

5) SW1 is trying to autonegotiate int speed settings with SW2. However autonegotiation is disabled on SW2 interface which is configure with speed of 100 mbps & ~~du~~ full duplex. What speed & duplex setting will SW1 use, assume it sensed speed correctly? Ans: Speed: 100 Mbps Duplex: Half

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